

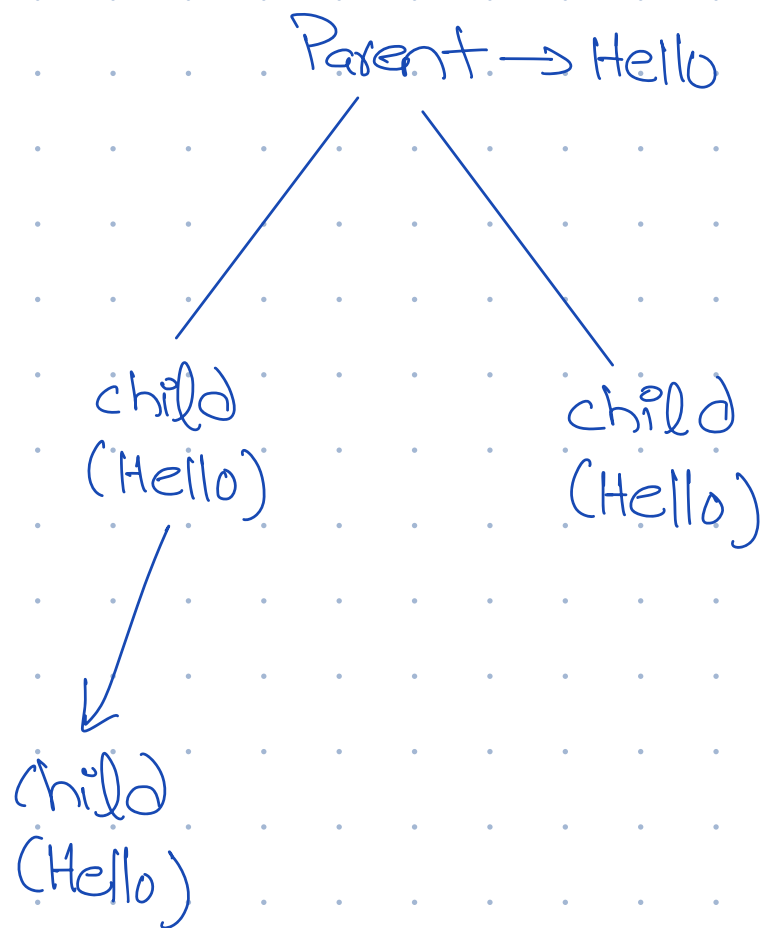
```
#include <stdio.h>
#include <unistd.h>

int main() {
    int x = 5;
    if (fork() == 0) { ← PC
        x += 5;
        printf("Child: x = %d\n", x);
    } else {
        x -= 5;
        printf("Parent: x = %d\n", x);
    }
    return 0;
}
```

10  
0

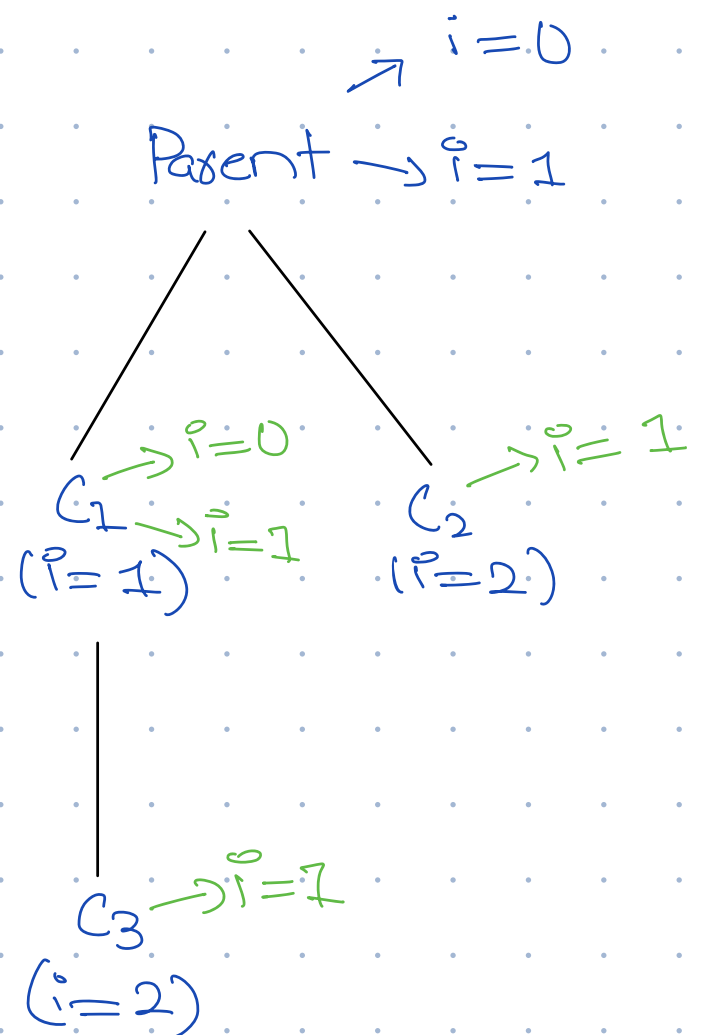
```
#include <stdio.h>
#include <unistd.h>

int main() {
    fork();
    fork(); ← PC
    printf("Hello\n");
    return 0;
}
```



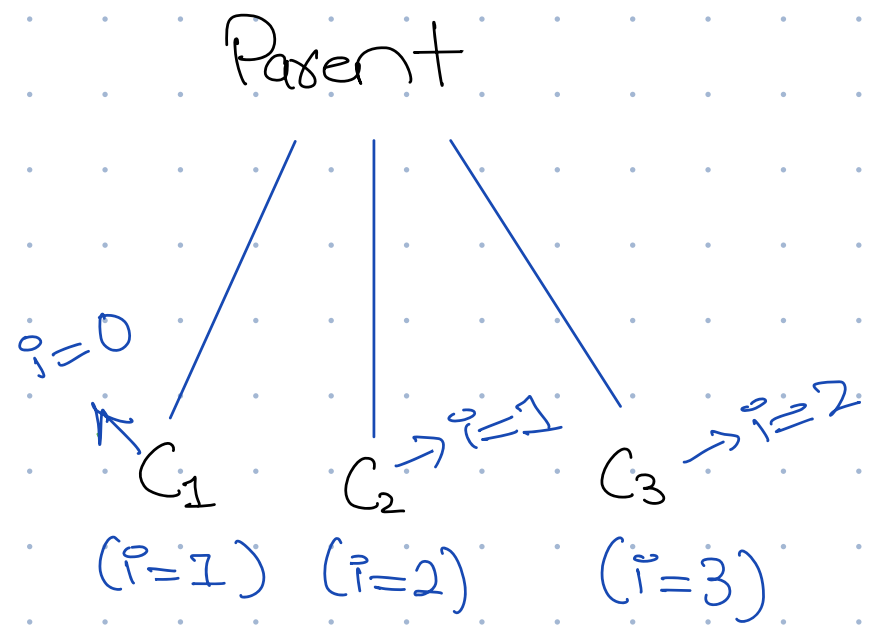
```
#include <stdio.h>
#include <unistd.h>

int main() {
    for (int i = 0; i < 2; i++) {
        fork();
        printf("i = %d\n", i);
    }
    return 0;
}
```



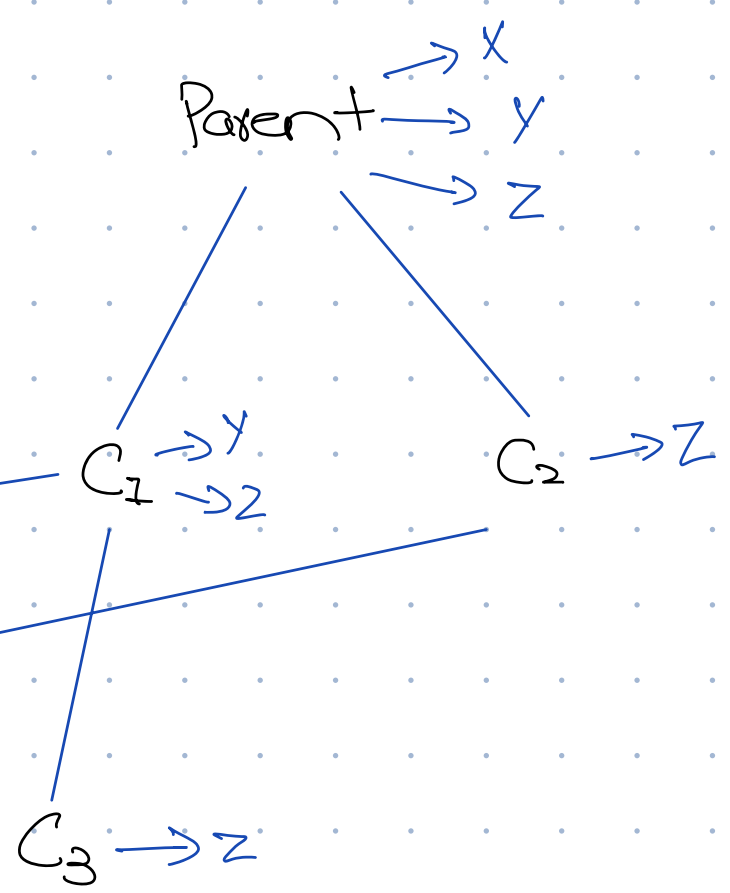
```
#include <stdio.h>
#include <unistd.h>

int main() {
    for (int i = 0; i < 3; i++) {
        if (fork() == 0) {
            printf("Child at i=%d\n", i);
            return 0;
        }
    }
    return 0;
}
```



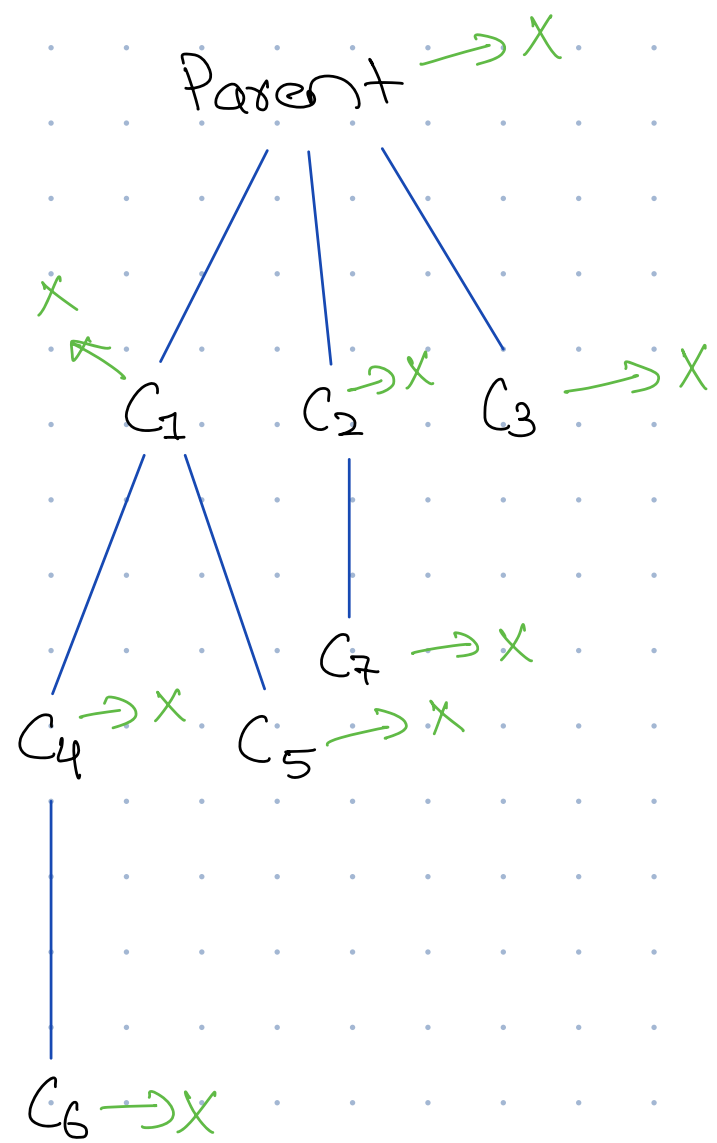
```
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("X\n");
    fork();
    printf("Y\n");
    fork();
    printf("Z\n");
    return 0;
}
```



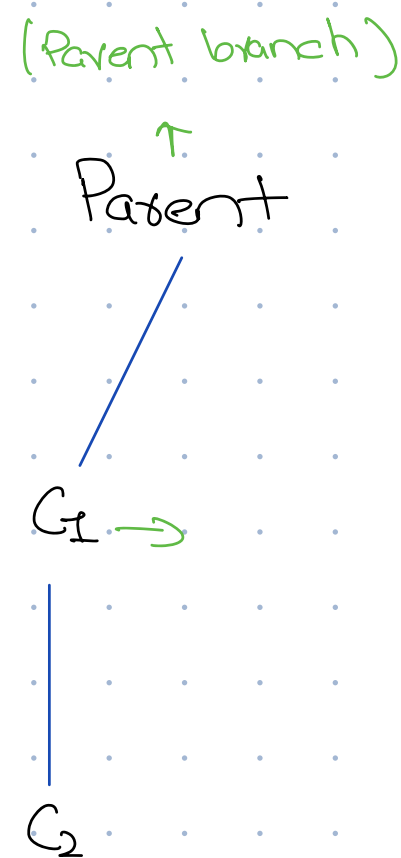
```
#include <stdio.h>
#include <unistd.h>
```

```
int main() {  
    fork();  
    fork();  
    fork();  
    printf("X\n");  
    return 0;  
}
```



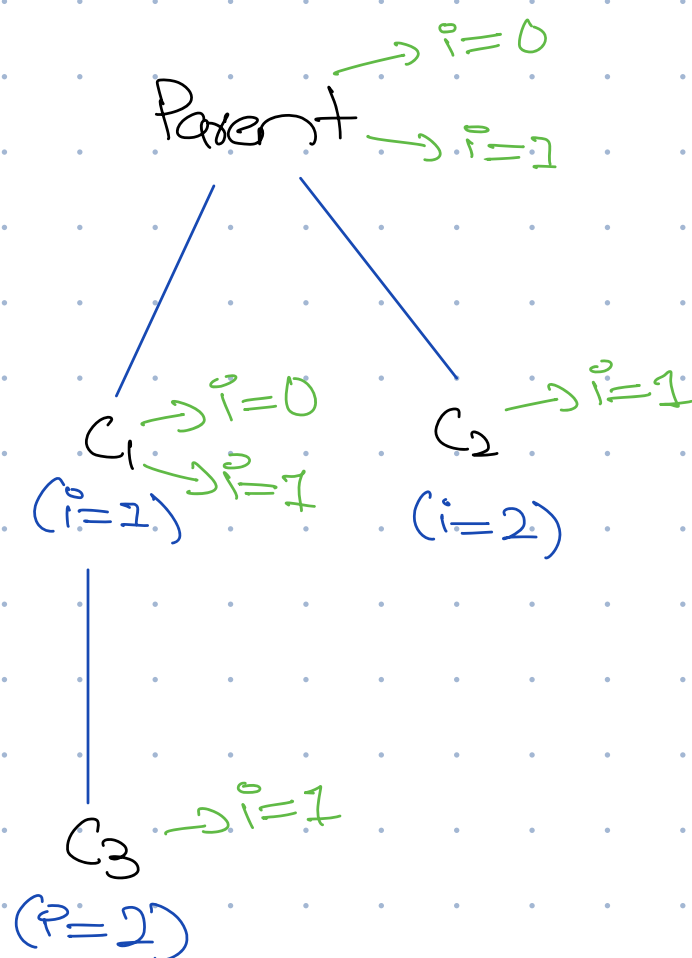
```
#include <stdio.h>
#include <unistd.h>
```

```
int main() {
    if (fork() == 0) {
        fork();
        printf("Child branch\n");
    } else {
        printf("Parent branch\n");
    }
    return 0;
}
```



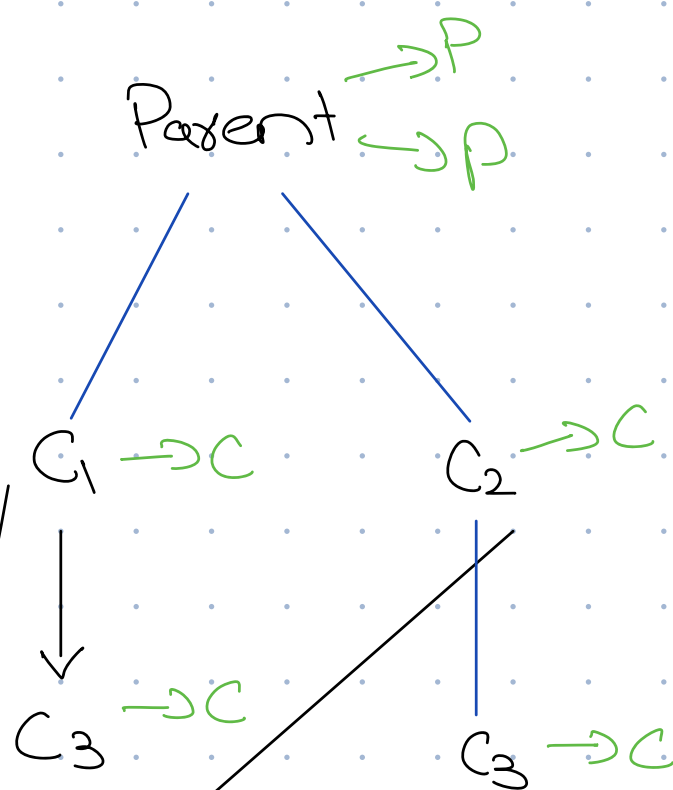
```
#include <stdio.h>
#include <unistd.h>

int main() {
    for (int i = 0; i < 2; i++) {
        fork();
        printf("i = %d\n", i);
    }
    return 0;
}
```



```
#include <stdio.h>
#include <unistd.h>

int main() {
    fork();
    if (fork() == 0) {
        fork();
        printf("C\n");
    } else {
        printf("P\n");
    }
    return 0;
}
```



fork()  
 rc = fork() ←  
 if (rc == 0) ←  
     fork()  
     printf("C")

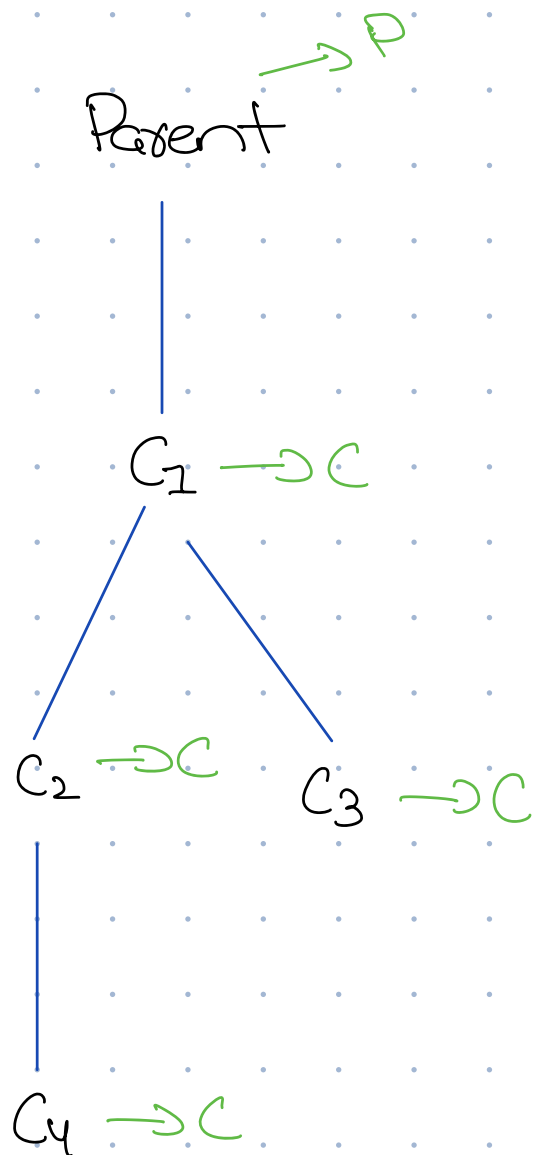
else  
     printf("P")

```

#include <stdio.h>
#include <unistd.h>

int main() {
    if (fork() == 0) {
        fork();
        fork();
        printf("C\n");
    } else {
        printf("P\n");
    }
}

```

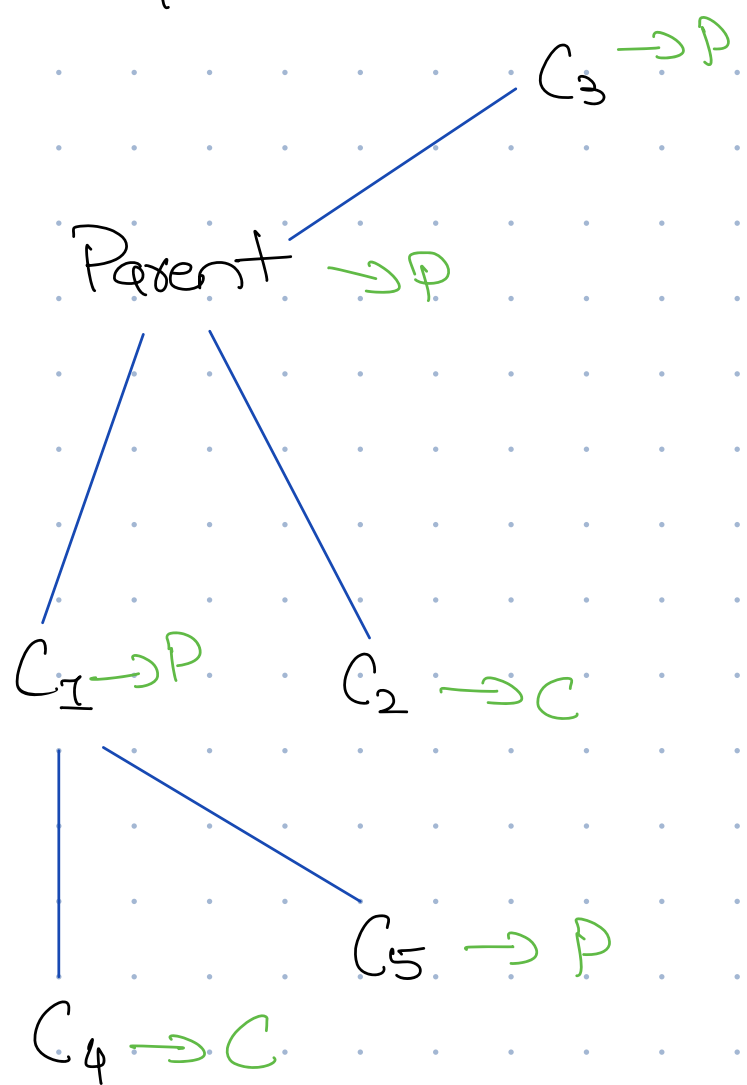


```

#include <stdio.h>
#include <unistd.h>

int main() {
    fork();
    if (fork() == 0) {
        printf("Child side\n");
    } else {
        fork();
        printf("Parent side\n");
    }
}

```



```
#include <stdio.h>
#include <unistd.h>
```

```
int main() {
    if (fork() == 0) {
        if (fork() == 0) {
            printf("Deepest child\n");
        }
        printf("Child branch\n");
    }
    printf("Everyone prints this\n");
}
```

Parent  
(Everyone)

C<sub>1</sub> (child branch)  
(Everyone)

C<sub>2</sub>

(Deepest child)  
(child branch)  
(Everyone)

Parent

C<sub>1</sub>

C<sub>2</sub>

child of child